Muhammad Talha Atif

Backend Engineer

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SUMMARY

Backend Engineer with 1+ year building RESTful APIs across Python, FastAPI, Node.js, NestJS, and Java Spring Boot. Worked with PostgreSQL, MySQL, MongoDB, and Redis, with hands-on experience in Kafka, RabbitMQ, and gRPC across distributed services. Shipped production deployments using Docker, AWS, and GitHub Actions CI/CD pipelines.

WORK EXPERIENCE

Full Stack Engineer Present
MicroTechX Pakistan (Remote)
Developed and maintained Java (Spring Boot) backend services for an enterprise IAM platform designing RESTful APIs, refactoring legacy monolithic modules into distributed microservices, and implementing event-driven workflows across 8+ services.

- Architected Kafka event-driven microservices for IAM audit trails across 8+ Spring Boot services configuring consumer concurrency, retry windows, and dead-letter queues with Maven.
- Implemented gRPC inter-service communication for user identity and manager resolution; refactored an O(n²) access-request lookup into an O(n) map-based strategy using Java Collections eliminating a critical performance bottleneck in the role-assignment approval flow.
- Enforced Spring Security-based authentication, JWT token validation, and RBAC access control across all service endpoints, wrote JUnit unit and integration tests for tile, cart, and request modules maintained GitHub Actions CI/CD pipelines for automated build, test, and deployment in Agile/Scrum sprint cycles.

Full Stack Engineer Sep '25 — Jan '26
Digital Upgraders LLC United Arab Emirates (Remote)
Built and maintained production features for a social networking platform serving photographers, videographers, and creative professionals across web and mobile.

- Developed RESTful Node.js APIs with layered service architecture, DTO-based input validation, and Supabase (PostgreSQL) ORM queries applying Object-Oriented design principles and backend patterns.
- Implemented rate limiting on authentication and high-frequency endpoints, JWT-based session management, and structured audit logging enforcing secure API access patterns across all service layers.
- Resolved 20+ production bugs spanning auth validation, notification lifecycle, cache invalidation, and protected route enforcement, maintained unit test coverage across critical API flows and validated fixes against multi-scenario edge cases before deployment

PROJECTS

Moodly- Experience Booking Platform Link Aug '25 — Present

- Built a FastAPI service that detects emotions from facial images and voice recordings using DeepFace and HuBERT models, returning a ranked confidence score across 7 emotion types consumed by the mobile app.
- Handled audio preprocessing end to end WAV file validation, stereo-to-mono conversion, and 16kHz resampling before feeding audio into the emotion classification model with structured JSON responses.
- Implemented file validation, automatic temp file cleanup, typed error responses, and a separate text embedding endpoint using SentenceTransformer keeping the API reliable and easy to integrate from the frontend.

Memingo - Language Learning Platform Link Sep '24 — Present

- Optimized leaderboard API performance by implementing Redis caching, reducing response time from 30 seconds to approximately 90ms and eliminating user drop-off to enable seamless real-time rankings
- Built a real-time object-detection module that identifies items and instantly delivers vocabulary, cultural context, and learning tips, transforming the app into a highly context-aware language learning experience
- Integrated a Gemini API-powered chatbot to provide personalized learning paths and resources, accelerating average learner progression

SKILLS

Programming Languages SQL, Java, Python, Javascript, TypeScript
Databases MySQL, MongoDB, SQ Lite, PostgreSQL, Supabase, MariaDB, Redis
Development Tools Android Studio, VS Code, IntelliJ IDEA, Postman, Eclipse, Pycharm
Version Control Github, Git
AI & Third Party Servcies Swagger, Stripe, Open Ai, Gemini, Hugging Face Models
Cloud Firebase, AWS, AWS S3, AWS RDS, Dokploy, Redis
DevOps CI/CD pipelines, Github Actions, Docker, Kubernetes
Messaging Apache Kafka, RabbitMQ, WebSockets

CERTIFICATIONS

Master Microservices with SpringBoot, Docker, Kubernetes, Udemy — May '25

AWARDS

Runner-up, App Development Hackathon (2025) — May '25
Organized by Punjab University

Participant, CALICO Berkeley Speed Coding Competition (2024) — Dec '24
University of California, Berkeley

Ranked, ICPC Asia Topi Online Programming Contest (2024) — Nov '24
Organized by International Collegiate Programming Contests

EDUCATION

Bachelor's degree in Computer Science, University of Management and Technology - UMT — Oct '21 — Aug '25
Lahore, Pakistan